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Machine Learning

Group Report

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AI-Driven Traffic Management: A Machine Learning Framework for Smart City Congestion

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1. Background & Case Study

1.1 Introduction to Urbanization and Smart City Technologies

The rapid urbanization of global populations has placed immense pressure on municipal infrastructure, making the mitigation of traffic congestion a pressing concern for emerging "Smart Cities". Smart cities refer to the implementation of advanced technological systems, including IoT (Internet of Things), AI (Artificial Intelligence), and ML (Machine Learning), for improved infrastructure development and sustainable living conditions. ITS (Intelligent Transport System) is one of the major elements of smart city technology that involves traffic management, mitigation of congestion, and improvement of road safety.

1.2 The Problem Statement: Urban Traffic Congestion

Urban traffic congestion remains a significant challenge for city planners, leading to increased travel times, widespread fuel wastage, and severe environmental pollution. Such congestion is mostly caused by the growing number of automobiles being used, as well as poorly coordinated traffic signals. Conventional methods of traffic control are static and reactive, in that they use traffic signals with a fixed time pattern, running through previously set cycles, irrespective of the situation at hand [1].

To counter the static nature of the conventional system, municipalities resort to extremely inefficient manual interventions [1]. It has become a necessity for traffic police officers to be deployed daily to specific, notorious, traffic-congested junctions in the morning and evening to help citizens navigate their commute to work and back home. Sometimes, controllers manually intervene by extending the duration of green light signals along routes that are heavily congested during peak morning periods. The only problem is that conventional systems are unable to revert automatically. Hence, the traffic signals get stuck along the prolonged morning route even during other off-peak periods.

Additionally, the excessive dependence on human labour is greatly hindered by the weather, which turns out to be unfavourable. With the onset of an unexpected rainstorm, which inevitably causes increased traffic volume as well as accidents, there is a delay in getting the traffic officers to their respective junctions, leading to the inability to control the congested roads when it is most necessary to do so [1]. In the absence of autonomous traffic management capabilities in such legacy systems, this is inevitable.

1.3 The Role of Machine Learning in Traffic Management

Machine learning provides a transformative approach to traffic management, enabling transportation systems to shift from reactive to proactive [2]. This will enable the system to become predictive to predict traffic flow, adaptive to automatically adapt to signal control, and efficient to reduce congestion and time spent traveling [2].

Out of all the different models that can be applied to this problem, ensemble-based models such as the Random Forest algorithm are very effective. Random Forest models are often used in ITS as they have the ability to analyse big data effectively, establish the complex nonlinear relationship between factors such as rush hour spikes in volume, and achieve great prediction accuracy. The prediction of traffic volume is vital in achieving optimization of traffic signals and long-term planning [1].

1.4 Case Study: Smart Traffic Volume Prediction System

To demonstrate the usage of ML in smart cities, this report presents a case study focused on developing a Smart Traffic Prediction System to estimate hourly traffic volume on a major highway.

1.4.1 Dataset Description

The "Metro Interstate Traffic Volume" dataset from the UCI Machine Learning Repository is used [3]. This dataset appears to be appropriate for analysing smart city projects as it includes both time and environment variables that represent the actual situation. Features used for building a predictive algorithm are:

a. **Time Variables**: date_time (data collecting timestamp).
b. **Environmental Variables**:
   i. temp (mean temperature in Kelvin)
   ii. rain_1h (volume of rain in the last hour in mm),
   iii. clouds_all (cloud coverage percentage)
c. **Target Variable**: traffic_volume (hourly traffic volume).

1.4.2 Exploratory Data Analysis

Before the model was trained, the raw dataset was studied by generating a statistical summary, plotting the distribution of the traffic volume, and generating a feature correlation heatmap to better understand the characteristics of the dataset. The statistical summary provides the exact count, mean, standard deviation, minimum and maximum value, and the 25th, 50th, and 75th percentiles of all features. Figure 1 shows that there are exactly 48204 rows of data, which is ample enough to train the Random Forest Regressor model. It can also be observed that the mean value for traffic volume is about 3260 cars.

	temp	rain_1h	snow_1h	clouds_all	traffic_volume
count	48204.000000	48204.000000	48204.000000	48204.000000	48204.000000
mean	281.205870	0.334264	0.000222	49.362231	3259.818355
std	13.338232	44.789133	0.008168	39.015750	1986.860670
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	272.160000	0.000000	0.000000	1.000000	1193.000000
50%	282.450000	0.000000	0.000000	64.000000	3380.000000
75%	291.806000	0.000000	0.000000	90.000000	4933.000000
max	310.070000	9831.300000	0.510000	100.000000	7280.000000

Figure 1: Statistical Summary of the Raw Dataset

Figure 2 shows the distribution of the hourly traffic volume feature in terms of the number of cars. It can be observed that there are three groups of traffic volumes with a high number of instances, which are around 500, 3000, and 4800 cars, with the first group having an exceptionally high number of instances compared to the other two groups. While the range of hourly traffic volume is rich, the distribution is not equally distributed across the whole range.

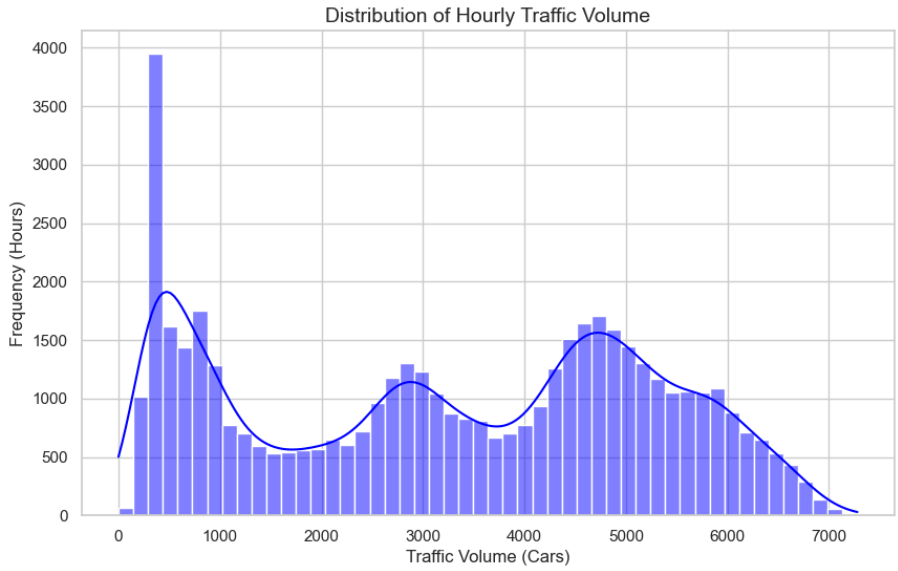

Figure 2: Distribution of Hourly Traffic Volume

Figure 3 shows the correlation between each feature to indicate the relationship between these features. It can be observed that there are almost no correlations between any of the features in the dataset, with the highest correlation coefficient being only 0.13. This is favourable for the ML model as highly correlated features can often provide very similar information, which can be redundant in the ML model. This is known as multicollinearity.

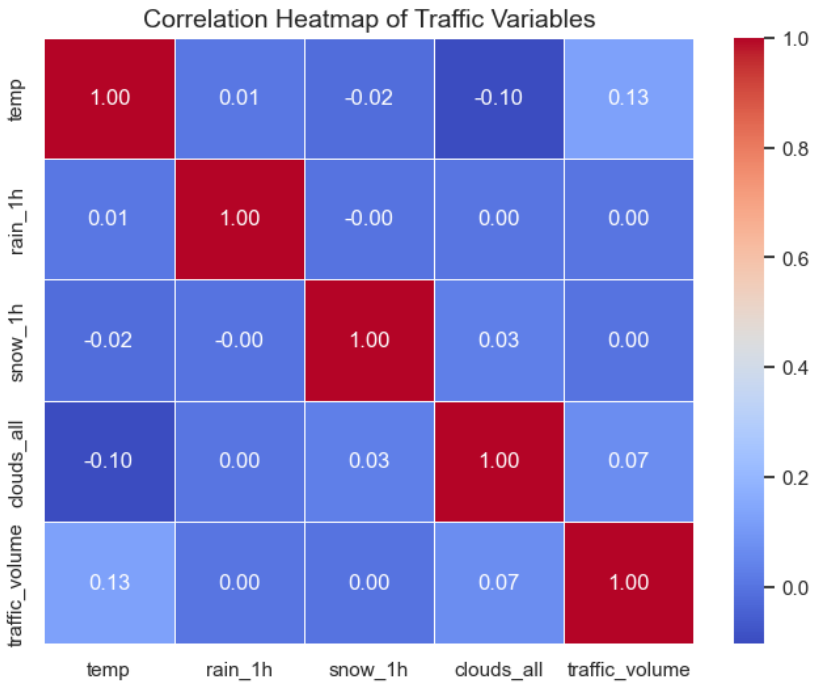

Figure 3: Correlation Heatmap of Dataset Features

1.4.3 System Innovation and Significance

The core innovative aspect of this system is that it successfully combines diverse and heterogeneous types of data, which are atmospheric information and temporal human activity, and condenses them into one prediction model. The use of the Random Forest algorithm allows us to identify non-linear dependencies that are impossible to determine by human operators.

For instance, we can calculate precisely when a combination of moderate rain and the evening traffic of working days leads to an exponential increase in congestion.

Rather than waiting for a bottleneck to form, this framework empowers municipal authorities to pre-emptively optimize traffic signal phases before peak hours begin [2]. This predictive capability represents the absolute forefront of ITS, directly translating raw environmental data into reduced municipal carbon emissions, optimized road usage, and seamless urban mobility.

1.5 IoT Sensors and Data Collection Process

To gather data from the real world, the ITS needs a reliable physical infrastructure for the IoT. The dataset used refers to physical IoT sensors deployed on highways [4].

a. Inductive loop detectors – Measure vehicle traffic count
b. Weather and Environmental Sensors:
    i. Temperature sensor – Measures ambient temperature
    ii. Rain gauge sensor - Measures rainfall amount per hour
    iii. Sky camera feed - Measures percentage of cloud cover

For the proposed traffic system, the gathered data is then sent to a central processing unit over 5G or fiber network connections. After data cleansing and temporal feature extraction, the data is then processed by a Random Forest machine learning algorithm to make predictions about traffic volume, allowing for proactive and efficient transportation planning.

2. Methodology and Flow Diagram of Proposed Model

2.1 Data Preprocessing

The dataset of the "Metro Interstate Traffic Volume" is fetched. Because standard machine learning algorithms cannot directly understand raw timestamps, the code performs feature engineering. It breaks down the `date_time` column into specific numerical components, which are `hour`, `day_of_week`, `month`, and an `is_weekend` binary flag. This step is critical because it translates human driving routines into a mathematical format that the ML model can process alongside environmental data (temperature, rain, snow, and clouds).

2.2 Model Training: Random Forest Regressor

To prevent data leakage and ensure the generalizability of the predictive framework, the dataset was partitioned utilizing a sequential cross-validation split. The data is split into 70% training set, 15% validation set, and 15% testing set.

The selected predictive engine is the **Random Forest Regressor**. Instead of following one decision-making route, the machine learning model uses an ensemble approach by creating 100 different decision trees using random partitions of the training set. Each decision tree predicts the number of vehicles, and the "forest" takes the average of these predictions. This strategy is extremely robust against anomalies such as a spontaneous traffic jam during the late-night shift.

Finally, the model assesses its performance based on R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Moreover, the Feature Importance score is calculated, which statistically identifies the importance of each input in the final prediction of traffic volume.

2.3 Actuation and System Workflow

Once the Random Forest model is trained and deployed to the cloud, it operates as the predictive engine within an Adaptive Traffic Control System (ATCS). The model ingests live IoT sensor data and outputs a forecasted traffic volume; the ATCS translates this prediction into immediate, physical interventions at the intersection level.

Rather than reacting to existing gridlock, the system pre-emptively executes the following signal adjustments before the predicted congestion arrives:

a. **Phase Lengthening (Green Light Extension):** In case the model is indicating an incoming influx of cars on the arterial road, the ATCS will automatically increase the period of time during which the light is green for that particular road, thus artificially boosting its traffic capacity.
b. **Phase Shortening (Red Light Management**): At the same time, the green light duration is shortened for the secondary intersecting side roads. By temporarily restricting access to the crossing road, traffic speed on the primary route is maintained.
c. **Dynamic Reversion**: As the model detects the conclusion of the peak commuting routine and predicts a return to low traffic volumes, the ATCS autonomously reverts the signals to shorter, baseline cycles. This ensures that off-peak drivers and pedestrians do not experience unnecessary delays at empty intersections.

2.4 Flow Diagram

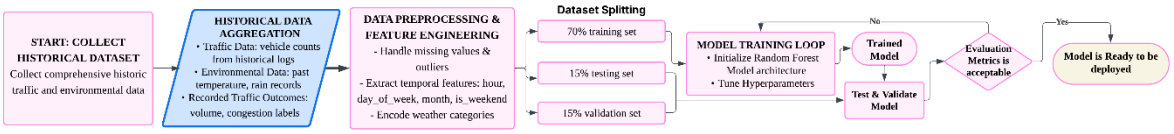

Figure 4: Flow Diagram of Model Training

Figure 4 above illustrates the workflow of training the model. The model is adjusted until the validation and testing result are in the acceptable range.

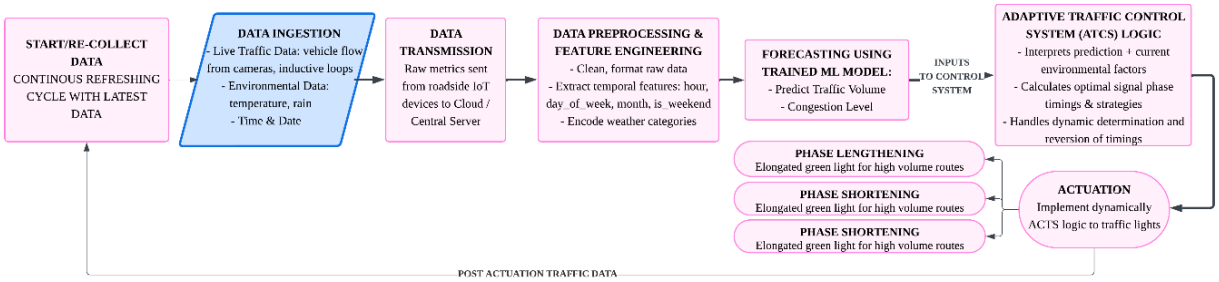

Figure 5: Flow Diagram of the Proposed Model

Figure 5 above illustrates the workflow of the trained model, demonstrating its readiness for real-world deployment in improving the efficiency of the Adaptive Traffic Control System (ACTS).

3. Result & Discussion

The performance of the Random Forest Regressor was evaluated using a multi-step verification process across the validation and test datasets. The model was trained on a 70% dataset, with 15% dataset for validation and the final 15% for testing.

3.1 Model Performance Metrics

The model achieved high predictive accuracy, as shown in Table 1, demonstrating its capability to handle the non-linear relationships between temporal patterns, environmental conditions, and traffic flow.

Metric	Validation Set	Test Set
R-squared (R^2)	0.9470	0.9568
Mean Absolute Error (MAE)	244 cars	234 cars
Root Mean Squared Error (RMSE)	459 cars	415 cars

Table 1: Evaluation Metrics

Figure 6 below show the correlation between the actual observed traffic volume and the volumes predicted during validation and testing.

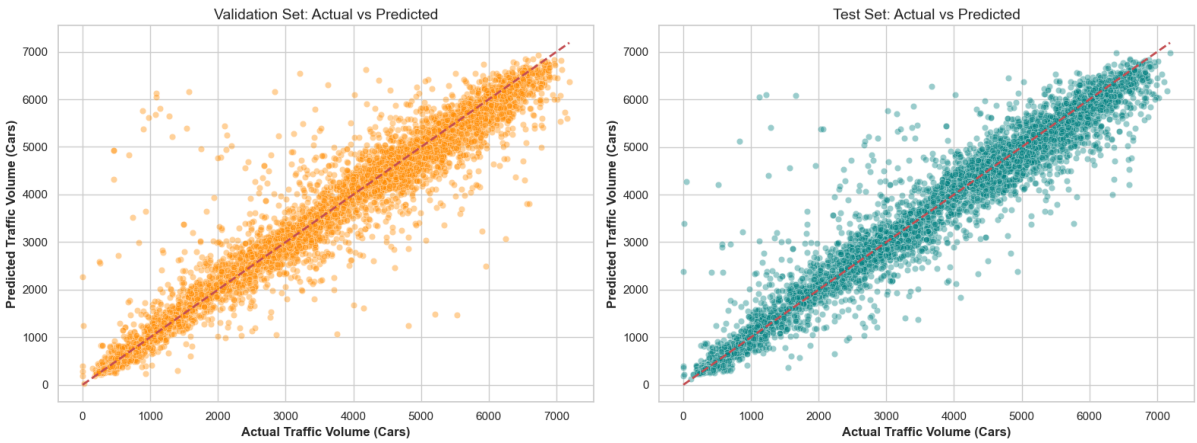

Figure 6: Predicted vs Actual plot

The strong concentration of data points along the red dashed 45-degree line indicates that the model's predictions are highly accurate across the entire range of traffic volumes. The high degree of similarity between the **Validation Set** and the **Test Set** visualizations confirms that the model is stable and robust against overfitting.

3.2 Feature Importance Analysis

Feature importance analysis was conducted to identify the key drivers of traffic volume, as shown in Figure 7. This statistical breakdown reveals which variables the model relies on most to make accurate forecasts.

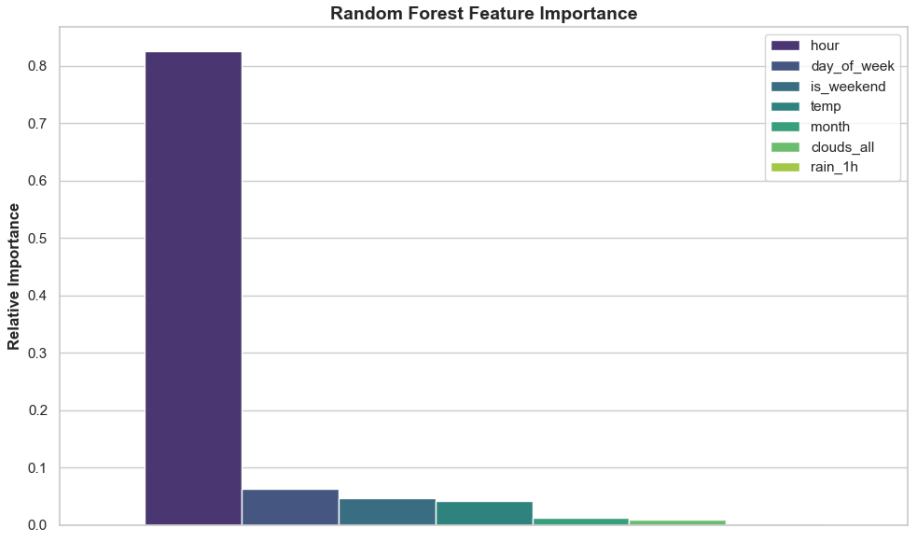

Figure 7: Feature Importance Score

The 'hour' variable that indicates the time of the day is the most dominant feature for over 80% of importance. Temporal indicators such as 'day_of_week' and 'is_weekend' were the next most influential, capturing the difference between work-day peaks and lighter weekend flows. Finally, environmental factors such as `temp`, `clouds_all`, and rain_1h, which denote temperature, percentage of cloud cover, and amount of rain per hour, showed measurable but lower relative importance compared to time variables. While weather impacts traffic, its effect is secondary to the established daily schedule of the city's inhabitants.

3.3 Discussion

Based on the model performance metrics, it can be concluded that the model is performing well with high accuracy. This is due to the high R^2 score achieved in both test and validation sets, 0.9568 and 0.9470, respectively. This means that the Random Forest Regressor model managed to predict 95.68% and 94.70% of the variance in the test and validation datasets, respectively. The error in predicting the traffic volume is very small, which can be considered negligible, depending on the city planner.

For the MAE values, there is a need to compare these values with the total number of cars on the road to gain useful insights. Based on the dataset, the number of cars on the road ranges from 0 to 7280 (peak rush hour), and the average is 3260. This means that the MAE of 234 represents an error rate of ~7.18% when the average traffic is 3260 cars, and ~3.2% when traffic is extremely bad at 7280 cars. It can be concluded that the error is small enough to be in the acceptable range.

Based on the feature importance score, it can be observed that the most influential feature that determines the traffic volume is the 'hour' feature, having a value of more than 0.8, and heavily overwhelms the other features. This shows that the state of traffic mostly depends on the institutional routines, such as office hours, school start times, and factory shifts. The congestion proves that they are more systematic rather than random. The day of the week and whether it is a weekend also have some influence on determining the traffic volume, as school and workdays are absent during the weekend. This will help the model better differentiate between the heavy, mandatory commuting of a Wednesday and the dispersed, lighter leisure travel of a Sunday.

Scenario	Input Features (Time/Weather)	Predicted Volume	ATCS Logic Decision
Morning Rush	8 AM, Weekday, Temp: 280K	5,420 cars	**Phase Lengthening** (Green +30s)
Rainy Afternoon	3 PM, Weekday, Rain: 5mm	4,150 cars	**Phase Lengthening** (Green +15s)
Off-Peak Night	11 PM, Weekday, Clear	840 cars	**Dynamic Reversion** (Baseline)

Table 2: ATCS Logic Decisions with Different Scenarios

There are multiple example scenarios to demonstrate the ATCS logic decisions. The first is the morning rush scenario at 8 AM on a weekday, where the temperature is 280K. The model predicted a traffic volume of 5420 cars, which is well above average, so the ATCS decides that it will execute phase lengthening. This highlights the model's proactive nature as the 'hour' feature strongly indicates a massive influx of commuters, so the ATCS directly adds 30 seconds to the green light on the main road without waiting for congestion to ensure smooth traffic flow.

The second scenario is the rainy afternoon situation at 3 PM on a weekday, with 5mm of rain. The model predicted a traffic volume of 4150 cars, which is just above average, so the ATCS decides that it will execute phase lengthening, but only adds 15 seconds to the green light. The predicted volume is above average despite not being during evening rush hour, but the model managed to calculate the wet roads and slow driving speeds introduced by moderate rain, so it altered the baseline predictions.

The third scenario is the off-peak nighttime at 11 PM on a weekday with clear skies. The model predicted a traffic volume of 840 cars, which is well below average, so the ATCS executes dynamic reversion to revert the traffic signals to shorter, baseline cycles, ensuring that the few drivers on the road do not experience unnecessary delays waiting at red lights for empty intersections.

4. Future Works

The constant evolution of the Smart Traffic Prediction System holds great potential for urban mobility. To better implement this in real life, future research must translate theoretical concepts into practical and testable frameworks, specifically focusing on **integration with edge computing** [5], **scaling for network interconnectivity** [6], and the use of **urban digital twins** [7].

For the topic of integration with edge computing, the current central cloud processing of IoT sensors data suffer from high latency and bandwidth limitations as they rely solely on centralized cloud computing to analyse this information. This can also result in a lack of local awareness and network congestion. To bring edge integration to reality, future systems must be deployed using a hybrid architecture. As an example, edge cache servers should be installed physically closer to IoT sensors at specific traffic "hotspots" to perform rapid, localized data filtering and real-time prediction, such as using fuzzy inference systems. The central cloud will then be reserved for permanent data storage and complex model training [5]. Future research papers can focus on testing this hybrid architecture by empirically comparing the response times, communication latency, and network bandwidth consumption of an edge-assisted model against a purely cloud-based system during simulated peak traffic hours.

In terms of scaling and network interconnectivity, many localized prediction models currently face limitations in overall scalability, as they only manage their own areas. To upscale this system, future researchers can develop hierarchical networks where multiple interconnected traffic hotspots communicate with a broader coverage area controller. This ensures that rerouting vehicles to ease congestion in one zone does not accidentally create another congestion in a neighbouring route. Future studies can implement this by using advanced models such as dynamic multi-graph neural networks (e.g., DMFGNet) [6], which simultaneously captures spatial and temporal traffic correlations across multiple regions. Researchers can then evaluate this by training the model on large-scale, citywide datasets to cover multiple areas and measuring the network-wide reduction in prediction errors.

For urban digital twins, implementing automated AI decisions directly into live physical traffic infrastructure carries high risks. To safely bridge this gap to reality, future research can rely on urban digital twins, which are virtual simulations of the city's physical infrastructure, using advanced, open-source traffic simulators like SUMO (Simulation of Urban Mobility). Importing real-world road networks into these platforms via mapping tools will allow researchers to create highly accurate virtual replicas of the city environment [7]. Future papers can test the machine learning framework’s effectiveness by simulating hypothetical scenarios, such as severe weather events like heavy rain or sudden road closures like a police officer roadblock and measuring the simulated Average Travel Time (ATT) of vehicles before and after the system intervenes. This virtual testing ensures the AI's automated routing decisions are reliable and safe before physical deployment.

5. Ethical Issues

The integration of ML into Smart Traffic Prediction Systems offers various benefits for urban mobility but also introduces ethical and security dilemmas. As highlighted by the Malaysian Ministry of Housing and Local Government's (MHLG) Smart City Framework, addressing these challenges is important to building sustainable and trustworthy smart cities, so that it improves the quality of life without compromising on safety. Future deployments of ML systems must carefully mitigate issues surrounding **data privacy violations**, **cybersecurity and public safety**, and **liability and reliability during system failures** [7] [8].

In terms of data privacy violations, smart traffic frameworks rely heavily on continuous collection of data from environmental sensors, loop detectors, and cameras, including images, proximity, temperature, etc. When these IoT devices unintentionally capture identifiable personal information, such as faces and vehicle license plates, it creates severe privacy risks [7]. The MHLG explicitly notes that robust information and data protection are necessary to secure stakeholder buy-in and public support. To maintain this trust, the MHLG framework stresses that any collected information must be protected and strictly used in accordance with the owners' wishes. This requires consistent reviews of personal data policies to achieve a sufficient balance between technological utility, transparency, and individual privacy.

Under cybersecurity and public safety, these AI models are prime targets for cyberattacks as they directly control critical physical infrastructure, such as traffic lights and automated routing systems. Malicious actors could exploit network vulnerabilities or launch data poisoning attacks to feed false traffic data to these sensors, reaching these AI models, which can potentially cause widespread gridlock or life-threatening accidents. By recognizing this severe risk to public safety, the MHLG framework identifies cybersecurity threats as a major urban challenge and establishes Policy 5 [8], which dictates that cybersecurity must be heavily strengthened within the smart city context. Securing these interconnected digital infrastructures with robust encryption and anomaly detection is of utmost importance to preventing malicious takeovers that could endanger civilian lives.

For liability and reliability during system failures, on the other hand, it can be observed that the reliance on automated AI systems raises complex ethical questions regarding these issues. Urban traffic systems must remain highly resilient and capable of reverting to fail-safe manual modes during power outages, hardware malfunctions, or network disruptions. If an AI-driven traffic controller fails or makes an inaccurate prediction that leads to a collision or delays emergency responders, it creates a massive legal and ethical dilemma [7]. Determining who is held responsible between the municipal government, the software developers, or the system operators is incredibly difficult, as all parties were involved in the implementation of the AI system. Resolving these liability and reliability concerns is important, as the fear of legal repercussions and public endangerment can severely thwart and delay the deployment of mission-critical AI systems.

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